PLSC 48110/28110

Lecture 8: Heterogeneous treatment effects and machine learning

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Loading packages for this class

```
> set.seed(60637)
> library(ggplot2)
```

Estimation of heterogeneous treatment effects

- Beyond questions about the ATE, we may have additional questions when running an experiment
  - Are specific subgroups, such as the most vulnerable subjects, affected differently by the intervention?
  - Is there a subgroup that is particularly helped by the intervention?
  - Is there a subgroup that is *harmed* by the intervention?
  - If we have a limited budget, how can we allocate treatment to maximize outcomes?
- We may know what subgroups we are interested in ex-ante.
- Or we may want to use data-driven methods to discover the subgroups most affected by our intervention.

When subgroups are pre-determined

- If you already know your subgroups, for example, students who are classified as female vs. otherwise
- And you just want to test a simple hypothesis of the form

$$\begin{aligned} H_0 : &\quad \tau(x) = \tau(x') \\ H_1 : &\quad \tau(x) \neq \tau(x') \end{aligned}$$

- You can compare simple differences in subgroup treatment effects.

$$\begin{aligned} &(\mathrm{E}\left[Y_i(1) - Y_i(0) | X_i = x\right]) - \\ &\left(\mathrm{E}\left[Y_i(1) - Y_i(0) | X_i = x'\right]\right) \end{aligned}$$

When subgroups are pre-determined

- With random assignment, we can just compare differences in differences in means across the two groups

$$\left(\mathrm{E}\left[Y_i^{obs}|W_i = 1, X_i = x\right] - \mathrm{E}\left[Y_i^{obs}|W_i = 0, X_i = x\right]\right) - \left(\mathrm{E}\left[Y_i^{obs}|W_i = 1, X_i = x'\right] - \mathrm{E}\left[Y_i^{obs}|W_i = 0, X_i = x'\right]\right)$$

- Where for each subgroup, you can estimate as the difference in average outcomes among individuals under treatment vs. control.

$$\hat{\tau}_x = \bar{Y}_{1,x}^{obs} - \bar{Y}_{0,x}^{obs}$$

When subgroups are pre-determined

- You can also use a linear model; this will return the same estimate

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 W_i + \hat{\beta}_2 W_i \times (X_i = x) + \hat{\beta}_3 (X_i = x)$$

A joint test of treatment-effect heterogeneity

For K prespecified, mutually exclusive subgroups, fit:

$$Y_i = \alpha + \tau W_i + \sum_{k=1}^{K-1} \delta_k G_{ik} + \sum_{k=1}^{K-1} \gamma_k W_i G_{ik} + \varepsilon_i.$$

- G_{ik} indicates subgroup membership; omit one reference group.
- Test the interaction coefficients jointly:

$$H_0 : \gamma_1 = \cdots = \gamma_{K-1} = 0.$$

- Use a robust Wald/F-test of these interactions, with covariance estimation appropriate to the assignment design.
- Rejection is evidence that some subgroup treatment effects differ. Identifying which ones requires follow-up comparisons and attention to multiple testing.

When subgroups are not pre-determined

- You might be interested instead in answering the question, "Is there any treatment effect heterogeneity?" or
- Which group benefits the most from the program?

- For a broader range of questions, we might want to be able to model

$$\tau(x) = \mathrm{E}\left[Y_i(1) - Y_i(0) | X = x\right]$$

- If all covariates are discrete and there are few of them, we can fit a fully interacted regression model, as before.
- But when there are a lot of covariates and/or they are continuous, we may want to have a more flexible estimation method.
- Machine learning methods can be helpful to address data-driven problems.

Machine learning

What is it?

- A body of *algorithmic* methods . . . (an algorithm is just a recipe)
- Somehow part of *artificial intelligence*. . . (basically, how computers perform tasks)
- In general, a flexible, *data-driven* approach to make predictions, classify data, or take decisions.

Machine learning

How do the objectives of machine learning differ from those of conventional quantitative methods in the social sciences?

- In conventional quantitative methods:
  - identify and estimate a target estimand, which is often a parameter in a statistical model, defined over some specified population of interest.
  - descriptive or causal, e.g., population employment rate, or effect of a policy
- In ML:
  - development of algorithms to make classifications or predictions.
  - e.g., is this a picture of banana or a cat? Will this person be more likely to click on an ad for sneakers or cookware?
- Is there overlap between the two?

Model fit vs. prediction

- In linear regression, propose a model:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \cdots + \hat{\beta}_K X_{Ki}$$

- Select $\hat{\beta}_0 \ldots \hat{\beta}_K$ to minimize

$$\sum_{i=1}^{N} \hat{\varepsilon_i}^2 = \sum_{i=1}^{N} \left(\hat{Y}_i - Y_i \right)^2$$

Model fit vs. prediction

- For prediction tasks, we could use the same model,

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \cdots + \hat{\beta}_K X_{Ki}$$

- But select $\hat{\beta}_0 \ldots \hat{\beta}_K$ to minimize squared prediction error for the next observation:

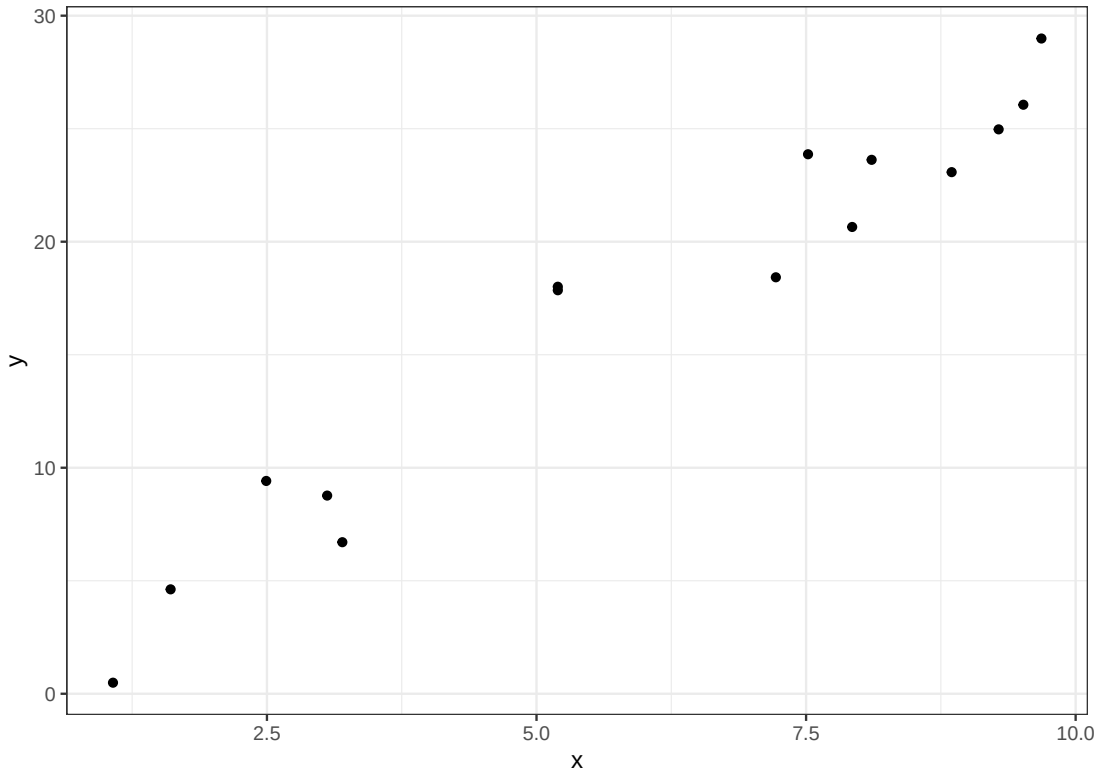
$$\hat{\varepsilon}^2_{N+1} = \left(\hat{Y}_{N+1} - Y_{N+1}\right)^2$$

- Are these the same thing? (no)
- If prediction is our goal, can we do better than least squares regression? (yes)

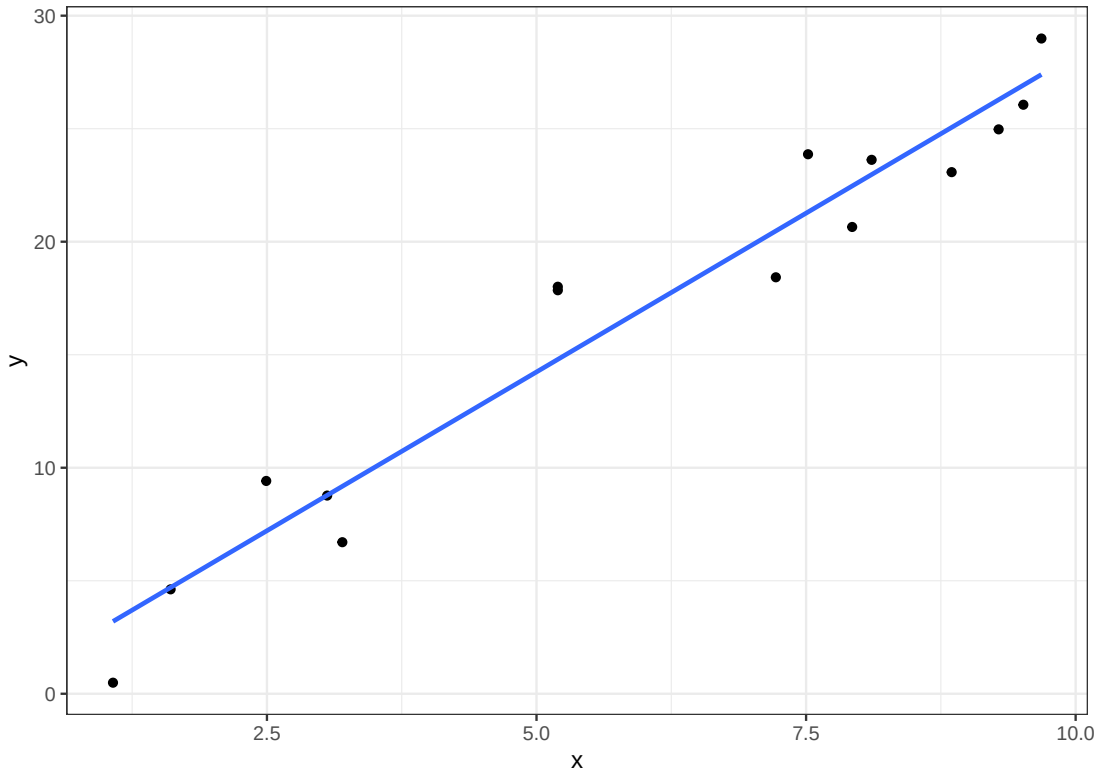
Some ML tools

- A major concern of ML: *overfit*
  - If your model fits the data *too* perfectly, it's not useful for prediction

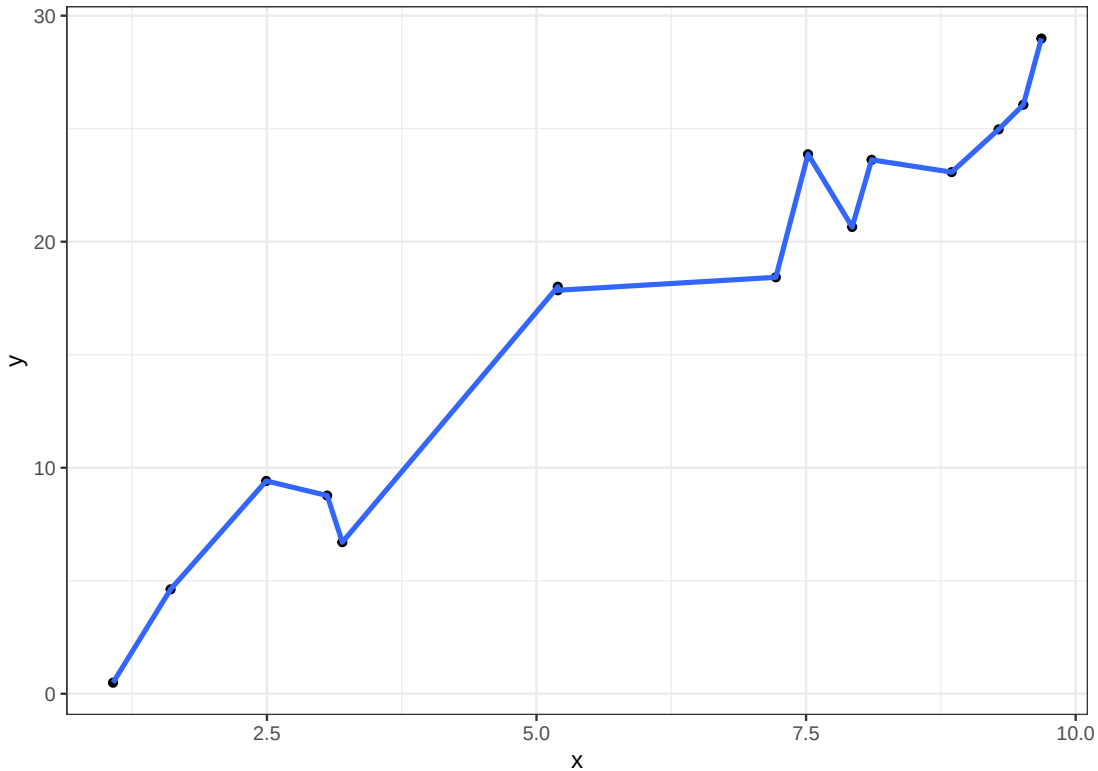
Suppose we would like to fit a model to the following data:



We could use a single line:



Or we could fit a line between every point:



Cross-validation

- If we were to draw another observation from the joint distribution of (Y, X), which one do you think would do a better job of prediction?
- ML methods propose a way to check this, by separating data into training and test sets.
- You can fit different models on the training set, and then see which one does the best job of predicting response in the test set. (This is not a new idea.)
- There are some different ways to do this:
  - Leave-k-out
  - Leave-one-out
  - k-fold cross validation

Regularization

- Overfit can become a real problem when we have a lot of predictors (K) relative to our number of observations (N)
- This is a common problem when we think about an industry setting, where for every customer a business might have a large number of measurements. Which ones should they use to predict an outcome?
- It is also a problem with predicting outcomes when we have a lot of complex covariates. Which covariates are important?

Regularization

- Consider a model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_K X_K + \varepsilon$$

- With $K \geq N$, even if every β_k is non-zero, we won't be able to make good predictions with all of our $\hat{\beta}_k$–and when we care about prediction, that's not our goal, anyhow.
- With regularization, we shrink some of the $\hat{\beta}_k$ nearly all the way or all the way to zero.
- For *ridge regression* or *lasso*, we select the $\hat{\beta}_k$ using:

$$\underset{\beta_0, \boldsymbol{\beta}}{\operatorname{argmin}} \left\{ \sum_{i=1}^{N} \left(Y_i - \beta_0 - \sum_{k=1}^{K} X_{ki} \beta_k \right)^2 + \lambda \sum_{k=1}^{K} |\beta_k|^q \right\}$$

Here $q = 1$ gives the lasso and $q = 2$ gives ridge regression.

Hastie et al. (2009)

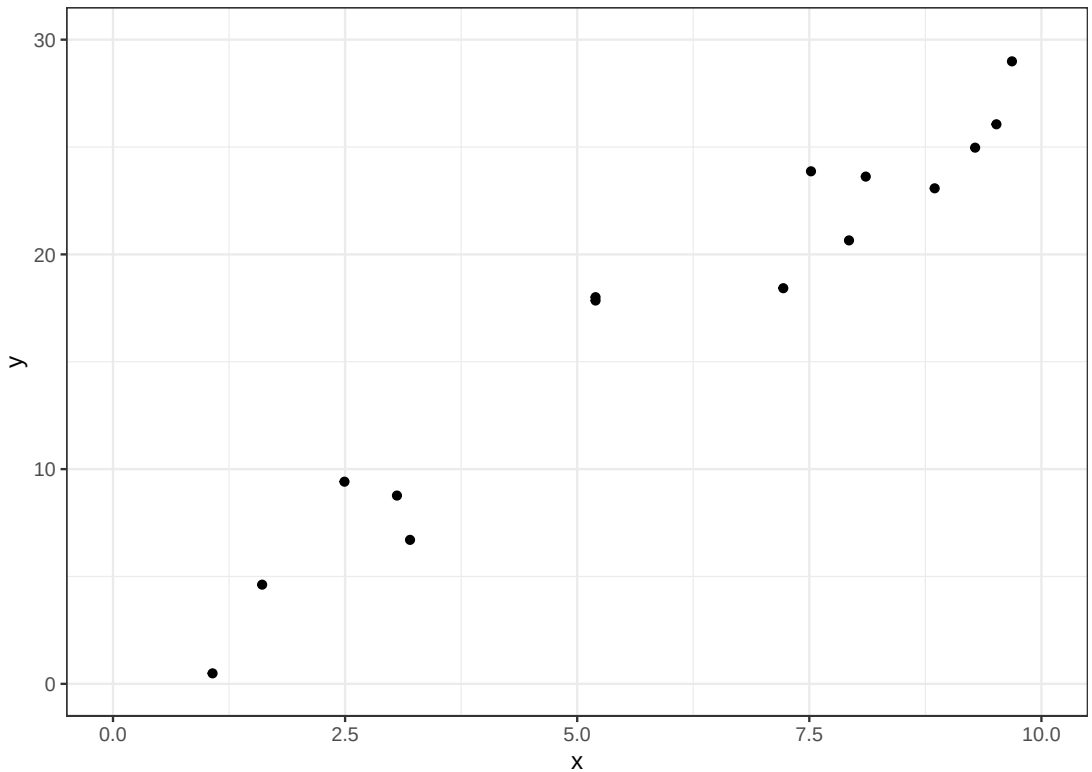
Regularization

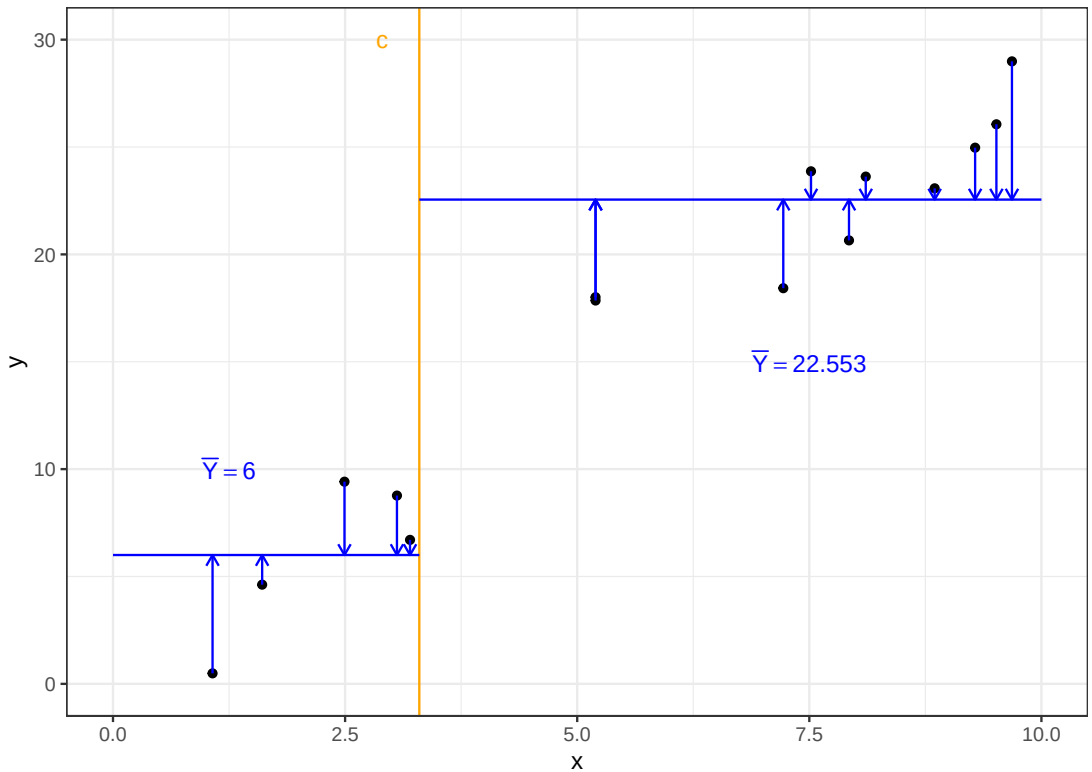
- For our model of $\tau(x)$, this means we fit something like a Lin estimator, to predict outcomes under treatment and under control, and determine which covariates are useful to include in our model.
- Bloniarz et al. (2016) suggest a procedure to do this for treatment effect estimation with the lasso.
- You could consider different penalties for different levels of interactions.

Regression Trees

- Suppose we have joint data, (Y, X), with just one predictor, X.
- Our goal is to pick some value of c so that we can split the data into two sub-samples ...
  - $X_i \leq c$
  - $X_i > c$
- ... and for each sub-sample, predict $\hat{Y}$ as the mean of the Y_i within each sample.
- We want to pick c to minimize:

$$Q = \sum_{i:X_i \leq c} (Y_i - \bar{Y}_{\text{lower}})^2 + \sum_{i:X_i > c} (Y_i - \bar{Y}_{\text{upper}})^2$$


30
20
10
0
y
0.0
2.5
5.0
7.5
10.0
x


c
$\overline{Y} = 6$
$\overline{Y} = 22.553$
y
x
0
10
20
30
0.0
2.5
5.0
7.5
10.0

Regression Trees

- Now suppose we have joint data, $(Y, X_1, \ldots, X_k)$.
- We will do the same approach to finding thresholds to minimize prediction error, but we'll want to pick which X_k we use for threshholding, as well.
- Generally, we'll define the depth of the tree as 2 or three variables; first we'll split on X_k, then we'll split on $X_j \ldots$

Regression Trees

- If we want to find a splitting point with respect to something other than mean response, such as treatment effects, we can target the treatment effect estimator instead of Y_i.

Honesty

- Returning to (causal) inference. . . we might like to use these methods to get valid inference, potentially on causal targets.

An honest tree algorithm

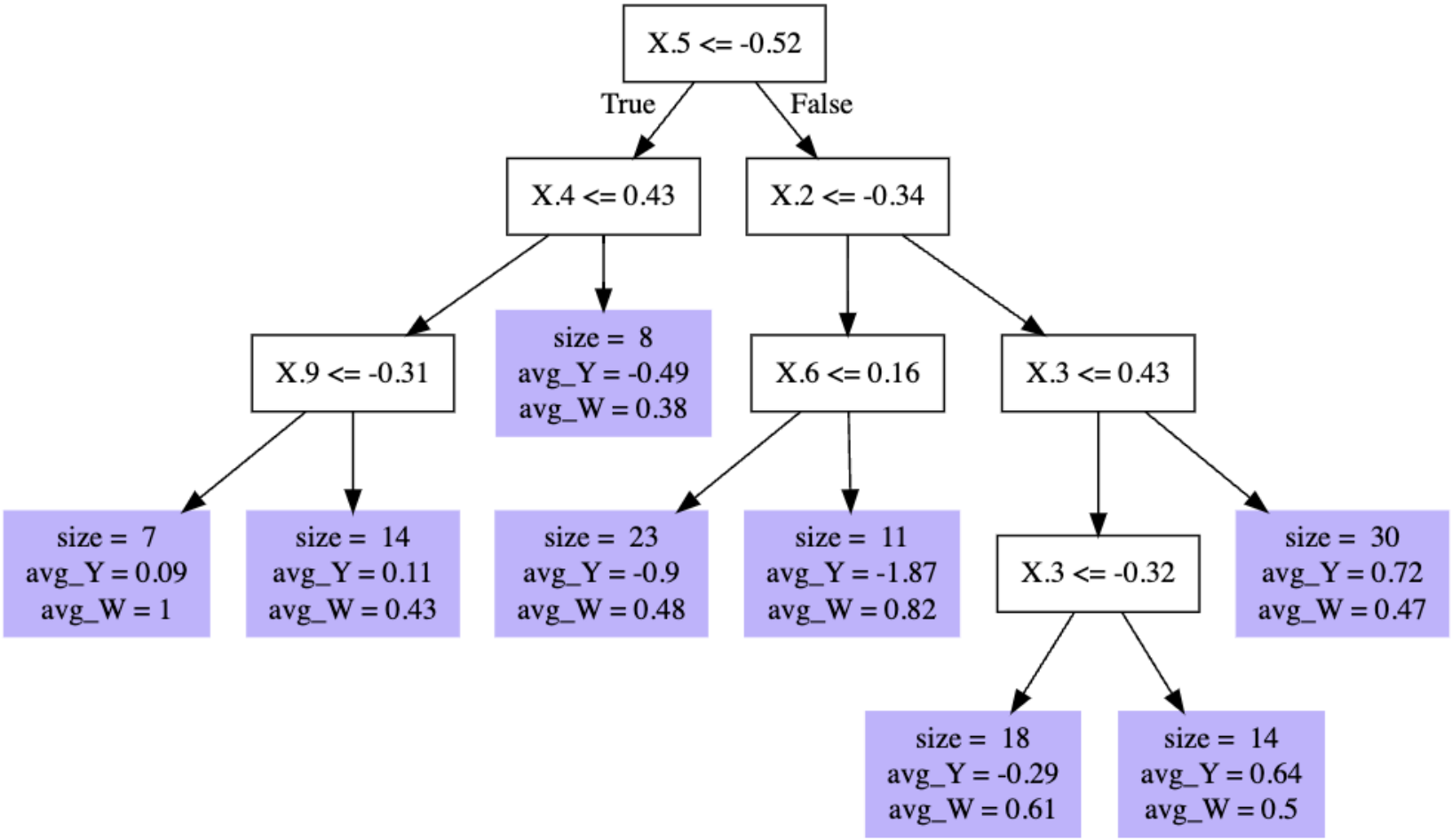
1. Split the sample into two folds.
2. Use the first fold to learn splits of the tree.
3. Estimate response within leaves using the second fold.

- This can result in some leaves being empty. Prune them?
- This procedure reduces bias relative to those proposed by Breiman (2001).

An honest tree algorithm

```
> library(grf)
> set.seed(60637)
> n ← 500
> p ← 10
> X ← matrix(rnorm(n * p), n, p)
> W ← rbinom(n, 1, 0.5)
> Y ← pmax(X[, 1], 0) * W + X[, 2] +
+ pmin(X[, 3], 0) + rnorm(n)
> c.forest ← causal_forest(X, Y, W)
```

An honest tree



Forests

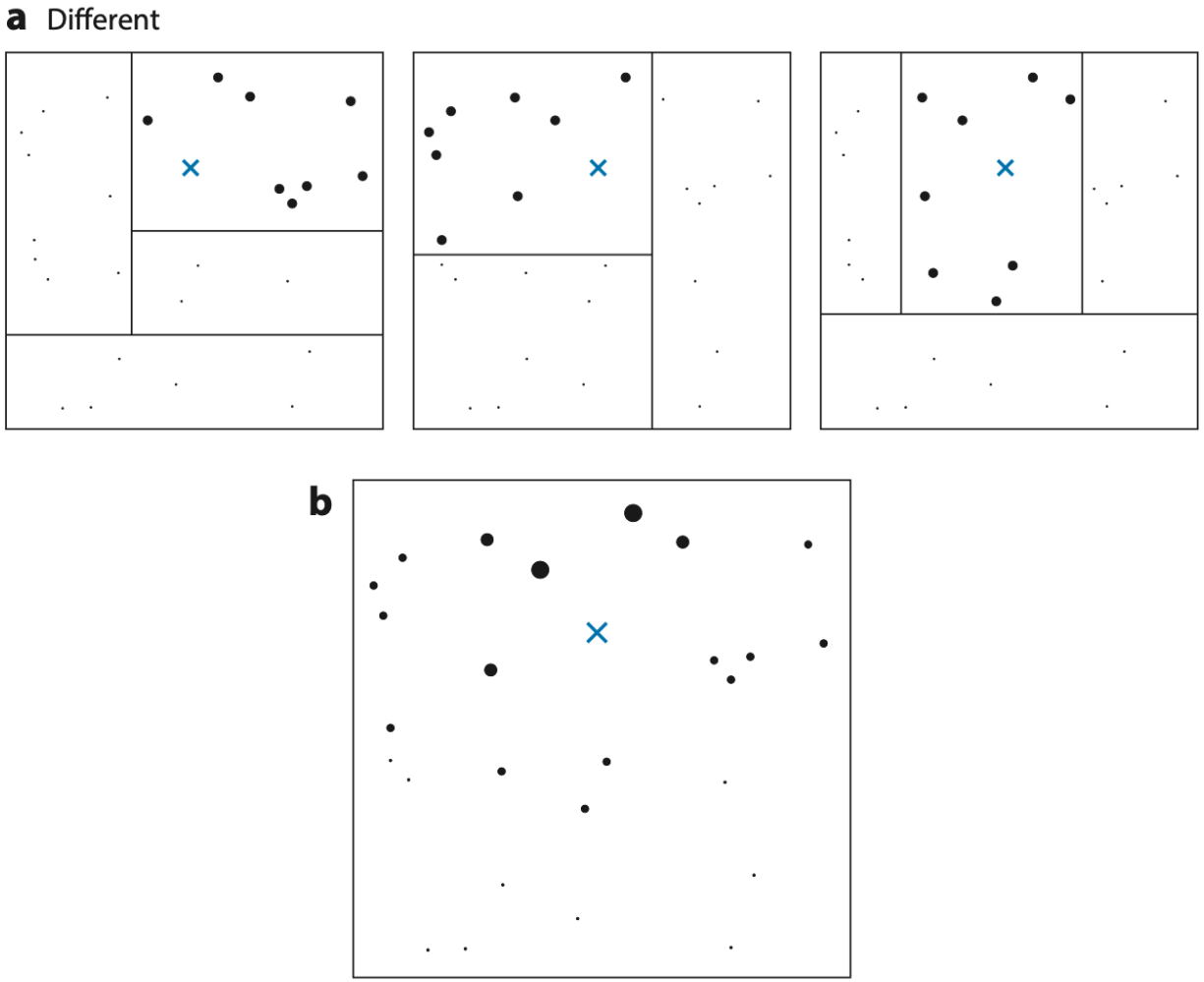


Figure 2

(*a*) Different trees in a random forest generating weights for test point X. (*b*) The kernel based on the share of trees in the same leaf as test point X.

Athey and Imbens (2019)

Policy learning

- Given information about an individual (age, location, education, etc.) which among multiple treatments should they be assigned?
- How do you learn a *policy* that will help you determine how to allocate treatment? . . . potentially accounting for different costs of treatment.

Matrix completion: the Netflix problem

- Netflix has data on viewers, their characteristics, and how they rate movies.
- The question: how to best recommend to them movies that they have not yet rated?
- The challenge: come up with the best recommendation algorithm, winner gets $1 million.
- This can be framed as a matrix completion problem: put users on rows, movies on columns, predict all of the missing rankings.

Causal inference: no free lunch

- Machine learning does not solve the fundamental problem of causal inference.
- Causal interpretations are based on assumptions about the data generating process, or knowledge of assignment procedures. These are outside the realm of machine learning methods.

Prediction error

- ML methods may do a good job of producing estimates, but how do we account for inference?
- Cross-validation
- Bootstrapping

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